

Increasing Hahn series cannot be evaluated at ω

A negative answer to Lipparini's Problem 7.7

20 September 2026

The stated problem. Let H be the ring of formal Hahn series

$$\sum_{\beta < \alpha} r_\beta x^{a_\beta}, \quad r_\beta \in \mathbb{R} \setminus \{0\}, \quad \gamma < \beta \implies a_\gamma < a_\beta,$$

with ordinal-length increasing surreal exponent supports, coefficientwise addition, and Hahn convolution. Problem 7.7 of [1] asks whether a ring homomorphism $\Phi : H \rightarrow \mathbf{No}$ can satisfy $\Phi(x^a) = \omega^a$ for every surreal a , with Conway's omega-map on the right.

Theorem. *No such homomorphism exists. In fact, no unital homomorphism from $\mathbb{Q}[[x]]$ into an ordered field can send x to an element greater than 1.*

Proof. The subring $\mathbb{Q}[[x]]$ is contained in H , since every subset of $\{0, 1, 2, \dots\}$ is well-ordered in the required increasing direction. Define $B(x) = \sum_{n \geq 0} b_n x^n$ by

$$b_0 = 1, \quad b_1 = -\frac{1}{2}, \quad b_n = -\frac{1}{2} \sum_{k=1}^{n-1} b_k b_{n-k} \quad (n \geq 2).$$

This is a well-defined formal series with rational coefficients. The constant and linear coefficients of B^2 are 1 and -1 . At every degree $n \geq 2$,

$$[x^n]B^2 = 2b_n + \sum_{k=1}^{n-1} b_k b_{n-k} = 0.$$

Consequently $B^2 = 1 - x$. If u is the image of x under a unital homomorphism, then

$$\Phi(B)^2 = \Phi(B^2) = \Phi(1 - x) = 1 - u.$$

For $u > 1$ this is a negative square, impossible in an ordered field. In the stated problem $u = \omega > 1$, and unitality follows from the prescribed image of x^0 . $\square$

Why this addresses ordinary ring homomorphisms. No step asserts $\Phi(\sum b_n x^n) = \sum b_n \Phi(x)^n$. The map is applied only to the finite algebraic identity $B^2 + x - 1 = 0$. Continuity, order preservation, preservation of infinite sums, and fixation of arbitrary real coefficients are not assumed. Proper-class issues are irrelevant to this witness.

Stronger consequence. For every positive integer n , the series $1 \pm nx$ is a square of a unit in $\mathbb{Q}[[x]]$, by the same recursion or substitution in B . Its image is strictly positive, so every unital homomorphism into an ordered field satisfies

$$-\frac{1}{n} < \Phi(x) < \frac{1}{n} \quad (n \geq 1).$$

Thus x must map to zero or an infinitesimal, never to ω . The accompanying article proves the converse for surreal targets and the classification of pure monomial prescriptions for general Hahn series.

[1] Paolo Lipparini, *Monotone infinitary operations on ordinals (extended version)*, arXiv:2505.00424v2, 30 April 2026, Problem 7.7, p. 38.

The question is explicit in the inspected revision. A focused search on 20 September 2026 did not locate a prior resolution; priority is not certified. This independent argument has not been refereed or proof-assistant checked.